6 **Ans: C**

Calculate the extra power needed for gravity

The force of gravity pulling the car backwards down the slope is the component of its weight parallel to the slope.

$F_{\text{down slope}} = mg \sin(\text{angle})$

$F_{\text{down slope}} = 1400 \times 9.81 \times \sin(2°)$

$F_{\text{down slope}} = 479.3 \text{ N}$

Power needed to overcome this gravity while moving at 25 m/s.

$P_g = Fv$
$P_g = 479.3 \times 25$
$P_g = 11982.5$
$P_g \approx 12$ kW

Calculate Total Power

$TP = P + P_g$
$TP = 30 + 12$
$TP = 42$ kW